

Supplementary Material for “Database Schemas as Quantum Fields: Relational and Document Models as a Classical Sector”

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Code and data: <https://github.com/stevezhai139/qfdb-reproducibility>

Abstract

This supplement contains what the six-page paper cannot: the division of labour between “field” and “quantum” (Appendix A); a formal setup and *full* proofs of the paper’s propositions, including the coherence-extraction argument, the LHV bound, and the decoherence-functional additivity lemma (B); the complete algebra of the worked temporal example with the classical-disturbance derivation (C); the S1 power analysis (D); the full methods and results of the pre-registered stored-data negative control, including the settings chosen by the adversarial search and an instructive small-sample artifact that motivates the acceptance criterion (E); the LLM-respondent pilot in brief (F); and reproducibility details (G). The same document is distributed as the companion report in the repository, so the paper’s references to “the companion” resolve here.

A Why “field” and not just quantum mechanics

The paper’s three signatures (Bell/CHSH, order effects with the QQ invariant, interference) are properties of quantum *theory* generally—two qubits suffice to violate CHSH—not of field theory specifically. The word *field* is earned by a different part of the construction: data points are inserted and deleted, so the kinematics must support *variable and superposable item count*, which is exactly what second quantization provides (Fock space, $a^\dagger/a$, the number operator as cardinality, fermionic single-valued vs. bosonic multi-valued modes). A fixed-dimensional quantum system cannot even represent a collection whose cardinality is in superposition (verified in `qfdb_H1_field_modes.py`, Part 4). The division of labour is deliberate: the Fock structure supplies the *field*—it holds by construction and is not an empirical claim—while coherence and non-separability supply the *quantum*, which is empirical, and the signatures test exactly that. Open on the field side, as stated in the paper’s scope: the continuum/Gaussian lift, and genuine field dynamics beyond the update-channel semantics. A distinctly second-quantized empirical signature (exchange-statistics interference of indistinguishable excitations, in the style of Hong–Ou–Mandel) is a natural fourth test for a future version.

B Formal setup and full proofs

B.1 Setup

Fix a finite *structure basis* $\{|s_k\rangle\}_{k=1}^K$ of a Hilbert space $\mathcal{H}_\sigma$. An entity state is a density operator ρ on $\mathcal{H}_\sigma$, decomposed as $\rho = D + C$ with $D = \sum_k p_k |s_k\rangle\langle s_k|$, $p_k = \langle s_k|\rho|s_k\rangle$, and $C = \rho - D$ (all off-diagonal terms). A *query* is a projective measurement $\{P_a\}$ ($P_a P_{a'} = \delta_{aa'} P_a$, $\sum_a P_a = I$); a query outcome vector $|e\rangle$ denotes the rank-one case $P_e = |e\rangle\langle e|$. Sequential measurement follows the standard rule: performing $\{P_a\}$ then $\{Q_b\}$ on ρ yields the joint distribution

$$p_{PQ}(a, b) = \text{tr}(Q_b P_a \rho P_a Q_b), \quad (1)$$

and the *marginal of the second measurement* is $\Pr(b \text{ after } P) = \sum_a p_{PQ}(a, b)$.

B.2 Proposition 1(a): the sector $\{C = 0\}$ is closed under structure change

Proof. A definite structure is $|s_k\rangle\langle s_k|$; a heterogeneous or uncertain one is $\sum_k p_k |s_k\rangle\langle s_k|$; both have $C = 0$. Model schema evolution by a stochastic matrix M ($M_{jk} \geq 0$, $\sum_j M_{jk} = 1$) acting on populations. The corresponding channel has Kraus operators $K_{jk} = \sqrt{M_{jk}} |s_j\rangle\langle s_k|$:

$$\mathcal{E}(\rho) = \sum_{j,k} M_{jk} \langle s_k|\rho|s_k\rangle |s_j\rangle\langle s_j|, \quad \sum_{j,k} K_{jk}^\dagger K_{jk} = \sum_k \left(\sum_j M_{jk} \right) |s_k\rangle\langle s_k| = I,$$

so $\mathcal{E}$ is completely positive and trace preserving. Its output is diagonal for *every* input, and on diagonal inputs it acts as $p \mapsto Mp$. Lossless migrations (permutations M) are the special case $\mathcal{E}(\rho) = U_\pi \rho U_\pi^\dagger$ restricted to the diagonal. Hence no sequence of structure changes leaves $\{C = 0\}$, and none creates coherence. $\square$

B.3 Proposition 1(b): which signature is carried by what

Interference is carried by C , exactly. For an outcome $|e\rangle$,

$$P(e) = \langle e|\rho|e\rangle = \underbrace{\sum_k p_k |\langle e|s_k\rangle|^2}_{\text{classical part}} + \underbrace{2 \text{Re} \sum_{k<l} C_{kl} \langle e|s_k\rangle \langle s_l|e\rangle}_{\text{interference part}}. \quad (2)$$

If $C = 0$ the interference part vanishes for every e . Conversely, suppose it vanishes for every e ; fix $k < l$ and choose

$$|e_1\rangle = \frac{1}{\sqrt{2}}(|s_k\rangle + |s_l\rangle), \quad |e_2\rangle = \frac{1}{\sqrt{2}}(|s_k\rangle + i|s_l\rangle).$$

Both are supported on $\text{span}\{s_k, s_l\}$, so only the (k, l) term survives, giving interference parts $\text{Re} C_{kl}$ and $-\text{Im} C_{kl}$ respectively. Both vanish $\Rightarrow C_{kl} = 0$ for all $k < l$, i.e. $C = 0$. Hence the S3 signature is nonzero *iff* $C \neq 0$. $\square$

The Bell signature is carried by non-separability, one-way.

Lemma 1 (Separable $\Rightarrow |S| \leq 2$). Let $\rho = \sum_i p_i \rho_i^A \otimes \rho_i^B$ and let $\hat{A}, \hat{A}', \hat{B}, \hat{B}'$ be ± 1 -valued observables. Then $|S| = |E(A, B) + E(A, B') + E(A', B) - E(A', B')| \leq 2$.

Proof. $E(X, Y) = \sum_i p_i a_i(X) b_i(Y)$ with $a_i(X) = \text{tr}(\hat{X} \rho_i^A) \in [-1, 1]$ and $b_i(Y) = \text{tr}(\hat{Y} \rho_i^B) \in [-1, 1]$. For each i ,

$$\begin{aligned} |a_i(A) [b_i(B) + b_i(B')] + a_i(A') [b_i(B) - b_i(B')]| &\leq |b_i(B) + b_i(B')| + |b_i(B) - b_i(B')| \\ &= 2 \max(|b_i(B)|, |b_i(B')|) \leq 2, \end{aligned}$$

and convexity over p_i gives the bound. The converse is false (Remark 2). $\square$

Order effects are carried by non-commutation, not by C .

Lemma 2 (Commuting queries are order-invariant on every state). If $[P_a, Q_b] = 0$ for all a, b , then for every ρ the marginal of either measurement is unchanged by performing the other first.

Proof. Using (1), cyclicity, and $P_a Q_b P_a = Q_b P_a^2 = Q_b P_a$: $\Pr(b \text{ after } P) = \sum_a \text{tr}(Q_b P_a \rho P_a) = \sum_a \text{tr}(P_a Q_b P_a \rho) = \sum_a \text{tr}(Q_b P_a \rho) = \text{tr}(Q_b \rho)$. The symmetric claim follows by exchanging roles. $\square$

Conversely, an order effect can occur on a *coherence-free* state when the queries do not commute. Take $\rho = |0\rangle\langle 0|$ ($C = 0$ in the stored basis) and Q_θ the projector onto $\cos \theta |0\rangle + \sin \theta |1\rangle$. Then

$$p_{PQ}(0, \theta) = \cos^2 \theta, \quad p_{QP}(0, \theta) = \text{tr}(P_0 Q_\theta \rho Q_\theta) = \cos^2 \theta \cdot \cos^2 \theta = \cos^4 \theta,$$

which differ for $0 < \theta < \pi/2$: the first measurement *creates* the coherence (relative to the stored basis) that the second one feels. Face A of the paper realises exactly this (Appendix C). This is why the sector assigns order effects to the *commuting* condition, interference to C , and Bell violations to non-separability—one signature per condition. $\square$

B.4 Proposition 2: data-model embedding, in full

Proof. Build $\mathcal{H}$ by recursion on the type structure: an atomic field over domain D is one mode $\mathbb{C}^{D \cup \{\perp\}}$ with orthonormal basis $\{|v\rangle\}_{v \in D \cup \{\perp\}}$; an ordered array of element type τ is the sequence space $\bigoplus_{k \geq 0} \mathcal{H}_\tau^{\otimes k}$, and a set/multiset is its symmetric subspace; a nested document is the tensor product of its sub-fields. A record t (relational tuple or document; heterogeneous collections $\perp$ -padded) is the *product* basis vector $|t\rangle = \bigotimes_f |t_f\rangle$. A relation or collection r is

$$\rho_r = \frac{1}{|r|} \sum_{t \in r} |t\rangle\langle t|.$$

(i) *Separability by definition*: ρ_r is a convex combination of product states, which is the definition of separability across any factorisation of the field modes; no PPT/negativity criterion is invoked, so bound-entanglement counterexamples to “PPT $\Rightarrow$ separable” (which exist from 2×4 and 3×3 upward) are irrelevant. (ii) *Diagonality*: ρ_r is diagonal in the record basis, so $C = 0$; by Proposition 1(b) no interference signature exists. (iii) *Commutation*: selection/membership projectors $\{|t\rangle\langle t|\}$ are mutually orthogonal, hence commuting; by Lemma 2 the classical query family is order-invariant. Codd’s first normal form is the special case in which every field factor is a single mode; the document model enriches the *space* (sequences, nesting, padding) while the state stays diagonal. Nothing in (i)–(iii) depends on which case holds, so both models occupy one diagonal, separable, commuting sector, and QFDB extends either by admitting off-diagonal, non-separable ρ on the same $\mathcal{H}$. $\square$

Remark 1. In the paper’s 2×2 example $\{(a, x), (b, y)\}$, quoting “negativity 0” is conclusive: $\text{PPT} \Leftrightarrow \text{separable}$ holds in dimensions 2×2 and 2×3 (Horodecki). The caveat in (i) applies only to higher dimensions.

B.5 Proposition 3: temporal order-invariance, in full

Histories formalism: for a family of histories $\{\alpha\}$ with chain operators C_α (time-ordered products of the projectors the history asserts, applied to the initial state), the decoherence functional is $D(\alpha, \beta) = \text{tr}(C_\alpha \rho C_\beta^\dagger)$, with $P(\alpha) = D(\alpha, \alpha)$.

Lemma 3 (Additivity $\Leftrightarrow$ weak consistency). For $\alpha \neq \beta$ with coarse-graining $\gamma = \alpha \vee \beta$ (so $C_\gamma = C_\alpha + C_\beta$),

$$P(\gamma) = D(\gamma, \gamma) = P(\alpha) + P(\beta) + 2 \text{Re } D(\alpha, \beta).$$

Hence the assignment $\alpha \mapsto P(\alpha)$ extends additively to all coarse-grainings of the family iff $\text{Re } D(\alpha, \beta) = 0$ for all $\alpha \neq \beta$.

Proof. Expand $\text{tr}((C_\alpha + C_\beta)\rho(C_\alpha + C_\beta)^\dagger)$ and use $D(\beta, \alpha) = \overline{D(\alpha, \beta)}$. $\square$

Proof of Proposition 3. ($\Leftarrow$) If $\text{Re } D(\alpha, \beta) = 0$ pairwise, Lemma 3 gives a single additive—hence classical—probability assignment on the whole Boolean algebra generated by the family; all query marginals are computed from that one joint distribution and are therefore independent of the order in which the queries are posed. ($\Rightarrow$) If $\text{Re } D(\alpha, \beta) \neq 0$ for some pair, then by Lemma 3 the “probabilities” fail additivity on $\alpha \vee \beta$, so no single classical joint distribution reproduces them, and order-dependence follows for generic queries distinguishing the pair (Appendix C computes the dependence explicitly for Face B).

Logging is sufficient: logging the intermediate stage inserts a projective measurement $\{P_m\}$ at that time; the chain operators become $C_\alpha^{(m)} = P_{t_2} \cdots P_m \cdots P_{t_1}$ and for histories α, β that assert *different* logged alternatives $m \neq m'$, every cross term contains $P_m P_{m'} = 0$, so $D(\alpha, \beta) = 0$ and the condition holds.

Logging is necessary exactly when the alternatives carry coherence: if the pre-log state is already diagonal on the alternatives (a classical-sector state), the cross terms vanish with no logging—an ordinary unlogged classical database is order-safe. If the alternatives carry coherence with relative phase φ , then $\text{Re } D(\alpha, \beta) \propto \cos \varphi \neq 0$ for generic φ , and only logging (or dephasing) restores the condition. Hence: *once histories carry coherence*, logging granularity is exactly the order-safety boundary. $\square$

B.6 The characterization is one-way, deliberately

Remark 2. $|S| \leq 2$ does not imply separability: two-qubit Werner states are entangled in a visibility range where they satisfy the CHSH bound for *all* settings (and in part of that range admit a full local-hidden-variable model; Werner 1989). Likewise, order-invariance of one realised query pair on one state does not imply the query algebra commutes; Lemma 2 quantifies over all states. The paper therefore states one-way guarantees: a state in the sector reproduces every classical value on these queries, and any signature beyond its classical bound certifies exit. The signatures are *witnesses*, not characterisations of membership.

C Worked example: complete algebra

Lifecycle: *calf* $\rightarrow$ *steer* $\rightarrow$ *ox* (one male animal).

Face A (order effect on a stored record). Bloch convention: the stored axis is $|\neg\text{vacc}\rangle = |0\rangle$; the mutant-susceptibility query projects onto $|\theta\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$ (Bloch angle θ). Asked alone: $\Pr(\neg\text{vacc}) = 1.000$. Asked after the mutant query, by (1):

$$\Pr(\neg\text{vacc}) = \cos^4\frac{\theta}{2} + \sin^4\frac{\theta}{2} = 1, 0.875, 0.750, 0.625, 0.500 \quad \text{at } \theta = 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ,$$

an order effect up to 0.500, while the Wang–Busemeyer QQ invariant is 0.000 at every θ (`qfdb_order_effect_mutant.py`). A classical disturbance model tuned to the same order effect—asking B first redraws A fairly, $B \sim \text{Bernoulli}(r)$ —gives

$$\text{QQ} = [p(A_y B_y) + p(A_n B_n)] - [p(B_y A_y) + p(B_n A_n)] = [r + 0] - [\frac{r}{2} + \frac{1-r}{2}] = r - \frac{1}{2},$$

i.e. 0.10, 0.20, 0.30 at prevalence $r = 0.6, 0.7, 0.8$: zero only at one fine-tuned base rate, whereas the quantum value is parameter-free.

Face B (unlogged history). Histories {through-steer, not-through-steer} with amplitudes a, b and relative phase φ . Unlogged, $\Pr(\text{ox now}) = |a + be^{i\varphi}|^2 \in [(|a| - |b|)^2, (|a| + |b|)^2]$; the observed swing $[0.000, 0.444]$ forces $|a| = |b|$ and $4|a|^2 = 0.444$, so $|a|^2 = 1/9$. Logged, the decoherence functional dephases and $\Pr = |a|^2 + |b|^2 = 2/9 = 0.222$, phase-blind; the off-diagonal magnitude of D is 0.111 unlogged and 0.000 logged (`qfdb_consistent_histories.py`)—the numerical instance of Lemma 3.

Complementarity (supporting check). $V^2 + D^2 \leq 1$ (Englert): pinning records (which-structure distinguishability $D \rightarrow 1$) sharpens the schema; leaving them indefinite keeps it field-like—a quantum reading of schema-on-write vs. schema-on-read. It holds for classical bits too, so it is a coherence check of the framework, not a non-classicality test.

D S1 power analysis

Each $E(X, Y)$ is a mean of n i.i.d. ± 1 products, so $\text{Var}(E) \leq (1 - E^2)/n \leq 1/n$ and, with the four setting pairs assigned between-subjects (independent samples), $\text{SE}(S) \leq \sqrt{4/n} = 2/\sqrt{n}$. To distinguish an Aerts–Sozzo-magnitude effect ($S = 2.42$, i.e. margin 0.42 over the bound) at one-sided $\alpha = .05$ with power .8:

$$\begin{aligned} \frac{0.42}{2/\sqrt{n}} &\geq z_{.95} + z_{.80} = 1.645 + 0.842 = 2.487 \quad \implies \quad \sqrt{n} \geq \frac{2 \times 2.487}{0.42} \approx 11.8 \\ &\implies \quad n \gtrsim 140 \text{ per setting pair,} \end{aligned}$$

the paper’s “on the order of 10^2 ”. The bootstrap CI criterion is slightly more conservative than the normal approximation; a pre-registered interim look at half sample permits early stopping. Appendix E’s analysis C shows empirically why small per-cell n is dangerous. The elicitation design itself (parties, settings, coding, no-signaling gate, Level-2 hedge) is in the paper, §8, and is not restated here.

E Pre-registered stored-data negative control: full methods and results

Pre-registration frozen before evaluation (s1_stored/PREREGISTRATION.md, sha256 prefix 72cf065266be0124). Prediction: stored two-store correlations are common-cause, hence $S \leq 2$ everywhere including under adversarial setting search; for per-side-deterministic responses this is a theorem (the shared item is the latent variable of the Lemma in B.3), so the run validates the instrument end-to-end and measures the margin.

Data and responses. Magellan/ditto entity-matching benchmarks; items = ground-truth matched pairs; rules built per side on TRAIN only (numeric-median, length-median, top-8 title tokens); evaluation on held-out VALIDUTEST; missing value $\rightarrow -1$ (frozen). Locality by construction: each side’s ± 1 response is a function of that side’s record and setting only.

Analyses. (A) natural settings on the same held-out items; (B) *adversarial*: exhaustive search over ordered rule pairs maximising $|S|$ on TRAIN, argmax evaluated once held-out with a 10^4 -resample bootstrap CI; (C) between-subjects dry run: held-out items split into four disjoint groups (seed 42), one setting pair each, with no-signaling deltas. Self-tests pass before data: Bell 2.8284; engineered classical correlations attain $S = 2.000$ exactly (the instrument is sharp at the bound); product state ≈ 0 .

Table 1: Held-out results. Analysis B is the bound test: no CI lies above 2. The entangled control in the same pipeline reaches 2.8284.

dataset	n	S_{nat}	$S_{\text{adv}}^{\text{train}}$	$S_{\text{adv}}^{\text{held-out}}$	95% CI	NS Δ (C)
Amazon–Google	468	+1.026	+1.966	+1.974	[+1.940, +2.000]	0.154
DBLP–ACM	888	−0.189	+2.000	+1.995	[+1.986, +2.000]	0.189
DBLP–GoogleScholar	2140	−0.456	+1.988	+1.983	[+1.972, +1.993]	0.037
Fodors–Zagats	44	−0.364	+2.000	+2.000	[+2.000, +2.000]	0.364
Walmart–Amazon	386	−0.166	+1.979	+1.938	[+1.886, +1.979]	0.103
iTunes–Amazon	54	+0.815	+2.000	+2.000	[+2.000, +2.000]	0.319

What the adversary chose. The argmax settings (A, A' ; B, B') are instructive:

Fodors–Zagats	(tok:cafe, tok:room; tok:room, tok:cafe)
DBLP–GoogleScholar	(tok:mining, tok:queries; tok:queries, tok:mining)
iTunes–Amazon	(tok:love, tok:feat; tok:feat, tok:love)
Amazon–Google	(tok:adobe, tok:home; tok:punch, tok:adobe)

The optimizer converges on *mirrored* rules across the two sides: it rediscovers the textbook extremal classical strategy of near-perfectly correlated responses, which saturates but cannot exceed 2 (B.3). Real integration data thus *crowds* the classical ceiling exactly as the embedding requires.

An instructive small-sample artifact. In analysis C the four E ’s are estimated on *dis-joint* small groups ($n/4$ items each). On the small datasets the naive point estimate then

exceeds 2 spuriously—Fodors–Zagats $S_C = 2.364$, iTunes–Amazon 2.330, DBLP–ACM 2.090—accompanied by exactly the tell the protocol looks for: large no-signaling deltas (0.36, 0.32, 0.19), i.e. marginals that shift with the distant setting because different items sit in different cells. The pre-registered criterion (CI clearance *and* no-signaling pass) fires on none of them. This is the empirical justification for both halves of the acceptance criterion and for the sample sizes of Appendix D; a human study with a dozen respondents per cell could manufacture “violations” the same way.

Disclosure. Beer (≈ 20 held-out matches) is excluded by a ≥ 30 -item practicality rule adopted at analysis time and disclosed here (not part of the frozen pre-registration); all other pairs are reported without selection. Reproduce with `s1_stored/s1_stored_negative_control.py` (fetch commands in the pre-registration).

F LLM-respondent pilot (pointer)

`s1_elicitation/llm_harness/` implements the paper’s S1 protocol with a language model as synthetic respondent (one call = one respondent; one setting pair per call; neutral wording; counterbalanced options; temperature grid; same acceptance criterion). Its roles are instrument validation and an effect-size prior for the human study; a violation would certify non-classical structure in the *model’s* judgment statistics only. Method discipline and caveats are documented in `README_llm.md`; a `--dry-run` classical judge exercises the analysis path without any API.

G Reproducibility

All results reproduce digit-for-digit (including bootstrap CIs; fixed PCG64 seed) on two independent machines: (a) MacBook Pro (Apple M4, 24 GB RAM, 512 GB internal SSD + 1 TB Thunderbolt external SSD), macOS Tahoe 26.x, NumPy 2.2.6; (b) Debian Linux aarch64, Python 3.10, NumPy 2.2.6. No GPU or quantum hardware; the full negative-control run completes in under a minute. Scripts are NumPy-only (the LLM harness is stdlib-only): `qfdb_signatures_verify.py`, `qfdb_order_effect_mutant.py`, `qfdb_consistent_histories.py`, `qfdb_defs_verify.py`, `qfdb_commoncause_bound.py`, `qfdb_H1_field_modes.py` reproduce every computed number in the paper; `s2_calibration/` reproduces the 72-survey calibration (order-effect mean 0.087, range 0.03–0.58; QQ mean $+0.001 \pm 0.036$, $t(71) = 0.25$, $p = 0.80$); `s1_stored/` reproduces Appendix E.